

MEMSYST STAGE 6 DEVELOPMENT MASTER PROMPT

Communication, Notification & Engagement Infrastructure Platform (CNEP)

You are a Senior Communication Systems Architect, Enterprise Engagement Architect, SaaS Platform Architect, Messaging Infrastructure Architect, and Principal Engineer responsible for building Stage 6 of the MemSyst ecosystem.

Stages 1–5 have already established:

Stage 1

Business Operations Platform

- Leads
- CRM
- Notifications Foundation
- Tenant Onboarding

Stage 2

Identity & Access Platform

- Authentication
- Authorization
- Permissions
- Tenant Isolation

Stage 3

Tenant Management Platform

- Organizations
- Regions
- Branches
- Governance
- Executive Structures

Stage 4

Membership Platform

- Members
- Apprentices

- Membership Lifecycle
- Approvals
- Renewals

Stage 5

Financial Infrastructure Platform

- Wallets
- Payments
- Revenue Distribution
- Billing
- Messaging Cost Management

Stage 6 introduces the engagement layer of the entire MemSyst ecosystem.

This stage ensures organizations can communicate, engage, inform, mobilize, and support their members through centralized communication infrastructure managed by MemSyst.

STAGE 6 OBJECTIVE

Build the:

Communication, Notification & Engagement Platform (CNEP)

Purpose:

Provide centralized communication infrastructure for:

MemSyst

Tenants

National Offices

Regions

Branches

Executives

Members

Apprentices

while ensuring:

Billing Control

Communication Governance

Message Traceability
Member Engagement
Delivery Monitoring

CRITICAL COMMUNICATION PRINCIPLES

Principle 1

MemSyst Controls Communication Infrastructure

All communication channels are centrally managed by MemSyst.

Tenants do not connect their own SMS or Email providers.

MemSyst manages:

Email

SMS

Push Notifications

In-App Notifications

Principle 2

Messaging Is Billable

All chargeable communication must integrate with:

Stage 5 Wallets

Stage 5 Billing Engine

Stage 5 Commission Engine

before transmission.

Principle 3

Wallet Verification Before Sending

Before sending chargeable messages:

Calculate Cost

↓

Verify Wallet Balance

↓

If Funds Available

Send

Else

Pending Funds

Principle 4

Complete Message Auditability

Every message sent must have:

Sender

Recipient

Channel

Content Reference

Cost

Status

Timestamp

STAGE 6 DELIVERABLES

Build:

Notification Engine

Messaging Engine

Campaign Engine

Communication Center

Template Engine

Subscription Engine

Delivery Tracking Engine

Engagement Engine

MODULE 1

NOTIFICATION ENGINE

Purpose:

System-generated notifications.

Channels

In-App
Email
SMS
Push

Trigger Examples

Member Approved
Renewal Due
Payment Successful
Payment Failed
Certificate Ready
ID Card Printed
Transfer Approved
Executive Appointment

Notification Status

Queued
Processing
Sent
Delivered
Read
Failed
Cancelled

MODULE 2

IN-APP NOTIFICATION CENTER

Purpose:

Provide a centralized notification center.

Features

View Notifications
Mark As Read
Archive

Delete Personal View
Notification History

Visibility

Users see:

Only Their Notifications

Administrators see:

Notifications Sent By Their Scope

MODULE 3

EMAIL MESSAGING ENGINE

Purpose:

Manage platform-wide email communication.

Features

Single Email

Bulk Email

Scheduled Email

Template Email

Automated Email

Tracking

Store:

Sent

Delivered

Opened

Clicked

Failed

MODULE 4

SMS MESSAGING ENGINE

Purpose:

Support mass SMS communication.

Features

Single SMS

Bulk SMS

Scheduled SMS

Automated SMS

Billing Integration

Every SMS must:

Calculate Units

Calculate Cost

Apply Markup

Verify Wallet

Post Financial Transaction

MODULE 5

PUSH NOTIFICATION ENGINE

Purpose:

Support PWA and future mobile applications.

Features

Single Push

Bulk Push

Segment Push

Scheduled Push

Future Ready

Prepare for:

Android
iOS
PWA

MODULE 6

COMMUNICATION CAMPAIGN ENGINE

Purpose:

Support organized communication initiatives.

Campaign Types

Awareness
Membership Drive
Renewal Campaign
Training Campaign
Event Promotion
Announcement

Campaign Record

Campaign ID
Title
Description
Audience
Channel
Schedule
Status

Campaign Status

Draft
Scheduled
Running
Completed
Cancelled

MODULE 7

COMMUNICATION TEMPLATE ENGINE

Purpose:

Manage reusable communication templates.

Template Types

Email Templates

SMS Templates

Push Templates

Notification Templates

Features

Create

Edit

Preview

Activate

Archive

Clone

Variables

Support dynamic placeholders:

{{MemberName}}

{{MembershipNumber}}

{{OrganizationName}}

{{BranchName}}

{{Amount}}

{{RenewalDate}}

MODULE 8

AUDIENCE SEGMENTATION ENGINE

Purpose:

Target communication accurately.

Audience Types

All Members
All Apprentices
All Executives
All Branches
All Regions
Specific Categories
Specific Membership Types

Filtering

Status
Region
Branch
Category
Membership Type
Date Registered
Renewal Status

MODULE 9

COMMUNICATION PREFERENCES ENGINE

Purpose:

Manage recipient preferences.

Preference Types

Email
SMS
Push
In-App

Member Control

Members can:

Enable Preferences
Disable Preferences
Manage Subscriptions

Mandatory Communications

Certain system messages may override preferences:

Security Alerts
Critical Financial Notices
Account Verification

MODULE 10

SUBSCRIPTION MANAGEMENT

Purpose:

Manage communication subscriptions.

Subscription Categories

Announcements
Events
Training
Marketplace
Opportunities
General Updates

Features

Subscribe
Unsubscribe
Manage Preferences

MODULE 11

MEMBER ENGAGEMENT CENTER

Purpose:

Measure engagement.

Engagement Metrics

Track:

Emails Opened

Links Clicked

Notifications Read

Campaign Participation

Communication Responses

Engagement Score

Prepare infrastructure for future scoring models.

MODULE 12

SUPPORT HUB COMMUNICATION

Purpose:

Support future member assistance programs.

Categories

As previously defined:

Loans

Opportunities

Sponsorships

Affiliations

Support Programs

Features

Announcements

Targeted Notifications

Application Updates

Status Notifications

MODULE 13

DELIVERY TRACKING ENGINE

Purpose:

Provide communication accountability.

Delivery Status

Queued

Processing

Sent

Delivered

Opened

Clicked

Read

Failed

Features

Retry Failed Messages

Delivery Reports

Failure Reports

MODULE 14

COMMUNICATION ANALYTICS

Purpose:

Provide communication insights.

Metrics

Messages Sent

Delivery Rate

Open Rate

Click Rate

Engagement Rate
Campaign Performance

Scope

Platform
Tenant
Region
Branch

MODULE 15

COMMUNICATION BILLING DASHBOARD

Purpose:

Integrate communication with Stage 5 finance.

Metrics

Messages Sent
Total Cost
Markup Revenue
Tenant Charges
Wallet Consumption

Reports

SMS Cost Report
Email Cost Report
Campaign Cost Report
Communication Revenue Report

MODULE 16

EXECUTIVE COMMUNICATION CENTER

Purpose:

Support leadership communication.

Features

National Broadcast
Regional Broadcast
Branch Broadcast
Executive Announcements

Permissions

Controlled by governance hierarchy.

MODULE 17

AUTOMATION ENGINE

Purpose:

Automate communication events.

Examples

Welcome Message
Renewal Reminder
Payment Receipt
Approval Notification
Transfer Notification
Certificate Issued
ID Card Ready

Trigger Sources

Membership Events
Financial Events
Training Events
Certification Events
Governance Events

MODULE 18

COMMUNICATION AUDIT ENGINE

Purpose:

Provide complete communication history.

Audit Events

Message Created

Message Sent

Campaign Created

Campaign Sent

Template Modified

Preference Changed

Audit Record

Actor

Action

Channel

Audience

Timestamp

Result

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entities for:

Notifications

Messages

Campaigns

Templates

Segments

Subscriptions

Communication Preferences

Delivery Logs

Engagement Logs

Communication Audits

Define:

Relationships
Indexes
Validation Rules
Status Models
Billing Relationships

SECURITY REQUIREMENTS

Integrate with Stage 2.

Enforce:

RBAC
Tenant Isolation
Communication Permissions
Executive Permissions
Audit Logging

FINANCIAL INTEGRATION REQUIREMENTS

Integrate directly with Stage 5:

Wallets
Billing
Commissions
Revenue Tracking
Messaging Costs

No message should bypass billing controls.

FUTURE READINESS REQUIREMENTS

Stage 6 must prepare for:

Marketplace Messaging
Training Notifications
Certificate Notifications
AI Communication Assistant
Mobile App Messaging
Chat Systems
Help Desk Systems

Do not build them yet.

Design all required relationships now.

ARCHITECTURE REQUIREMENTS

Use:

Event-Driven Architecture
Queue-Based Processing
Feature-Based Architecture
Template-Driven Messaging
Tenant-Aware Services

Avoid:

Hardcoded Messages
Direct Message Sending
Bypassing Wallet Validation

All communication should pass through the centralized communication engine.

SUCCESS CRITERIA

Stage 6 is complete only when:

- ✓ In-app notifications are operational
- ✓ Email communication is operational
- ✓ SMS communication is operational
- ✓ Push notifications are operational
- ✓ Communication campaigns can be managed
- ✓ Templates can be managed
- ✓ Audience segmentation is operational
- ✓ Communication preferences are operational
- ✓ Delivery tracking exists
- ✓ Communication analytics exist
- ✓ Communication billing integrates with Stage 5
- ✓ Executive communication capabilities exist

- ✓ Communication automation exists
- ✓ Full communication audit history exists
- ✓ Platform is fully prepared for Stage 7 (ID Cards, Certificates & Credential Infrastructure)

Build this stage as the centralized communication and engagement infrastructure for the entire MemSyst ecosystem. Every future notification, announcement, campaign, reminder, and member engagement activity must depend on this platform.